Name: Muhammad Mahad Ashfaq

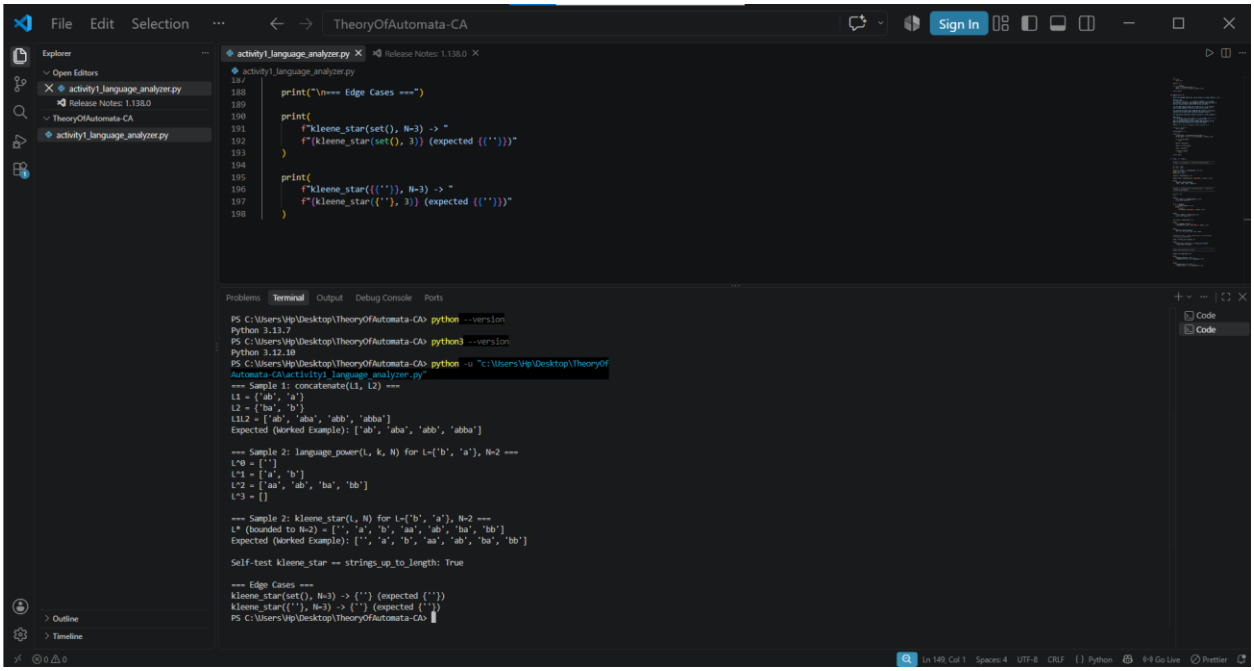
Roll no: BSSE24040

Class Activity 1

Submitted to: Dr. Muhammad Asif

Teaching Assistant: Muhammad Saqlain Tahir

1. Environment Verification and Program Output



2. What I Learned

The program is divided into separate functions so that each function performs one specific task, making the code easier to understand, test, and reuse. The concatenate() function is reused inside language_power() to construct higher powers of a language without repeating the concatenation logic. The bound N is used to remove strings longer than the required length, which keeps the computation finite even though language powers and the Kleene Star can grow indefinitely. The kleene_star() function repeatedly generates language powers and combines them until no new strings within the specified length bound are produced. The if __name__ == '__main__': section serves as the driver of the program by running the sample cases, self-test, and required edge cases. Overall, I learned how a small Python program can be organized into reusable functions and verified through test cases.

3. Confirmation

I successfully verified my Python installation using python --version and python3 --version. I also successfully ran activity1_language_analyzer.py in VS Code without errors, and the Sample 1, Sample 2, self-test, and required edge-case outputs matched the expected results.